



Science

Stage 8

Paper 2

2022

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	transports oxygen	1	Accept carries O ₂ Accept binds with oxygen
1(b)	increased numbers of cell B / increased number of white blood cells to destroy pathogens / to fight infection	2	each correct answer = 1 mark
1(c)(i)	plasma	1	
1(c)(ii)	transports (dissolved) nutrients / transports (dissolved) carbon dioxide	1	Accept transports (dissolved) urea/hormones/ amino acids

Question	Answer	Marks	Further Information
2(a)	A	1	more than one letter circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
2(b)	angle of incidence = angle of reflection / (angle) i = (angle) r	1	
2(c)	D	1	more than one letter circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence

Question	Answer	Marks	Further Information
3(a)	ink will dissolve in the solvent / pencil does not dissolve in the solvent	1	Accept solvent does not affect pencil / the ink will also be separated / the ink might run / the ink might wash off
3(b)	(idea that) different dyes move up the paper at different speeds	1	Accept higher level answers, such as colours have different R_f values / colours have different attraction to the paper
3(c)	yellow and green yellow and green spots match the dyes in the food colouring	2	both colours needed any order for the mark Accept spots travel the same distance Accept higher level answers, such as R_f values are the same

Question	Answer	Marks	Further Information
4(a)	25 (%)	1	
4(b)	fossil fuels	1	Accept non-renewable
4(c)	any two from wind tidal solar power hydrogen(-oxygen) fuel cell geothermal biomass / biofuel / biogas / bioethanol	2	each correct answer = 1 mark Accept wood/charcoal as biofuel
Question	Answer	Marks	Further Information
5(a)	(a pure substance) contains the substance only / no impurities	1	
5(b)	$\frac{10.8}{12.0} \times 100$ or 0.9×100 90 (%)	2	correct fraction / correct substitution = 1 mark correct answer (with or without working) = 2 marks

Question	Answer	Marks	Further Information															
6(a)	(to make it) a fair test	1	Accept (mass) is a control variable															
6(b)	(same) amount of water / (same) distance from flame	1	Accept same initial temperature Ignore same equipment															
6(c)	<table><tr><th>food</th><th>start temperature in °C</th><th>final temperature in °C</th></tr><tr><td>nut</td><td>28</td><td>64</td></tr><tr><td>biscuit</td><td>26</td><td>40</td></tr><tr><td>cereal</td><td>26</td><td>44</td></tr><tr><td>bread</td><td>27</td><td>38</td></tr></table>	food	start temperature in °C	final temperature in °C	nut	28	64	biscuit	26	40	cereal	26	44	bread	27	38	2	correct headings = 1 mark correct units = 1 mark Accept initial for start / end for final Note if no mark awarded accept 1 mark for three correct entries, e.g. two headings and one unit
food	start temperature in °C	final temperature in °C																
nut	28	64																
biscuit	26	40																
cereal	26	44																
bread	27	38																
6(d)	nut	1																
6(e)	fat / carbohydrate	1	Accept sugar(s) / starch / monosaccharides / saccharides															

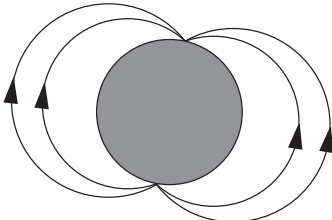
Question	Answer	Marks	Further Information
7(a)	(idea that) make sure north pole faces north pole (idea that) make sure south pole faces south pole	2	each correct answer = 1 mark ensure like poles face each other = 2 marks Accept marks from a labelled diagram
7(b)	(idea that) north pole faces south pole / unlike poles must face each other	1	Accept marks from a labelled diagram

Question	Answer	Marks	Further Information
8(a)	any two from (stellar) dust gas stars planetary systems	1	both answers correct for the mark Accept other objects found in galaxies Accept planets
8(b)	telescope	1	Accept named telescope, e.g. Hubble

Question	Answer	Marks	Further Information
9(a)	ecosystem	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
9(b)	any two from in the sand under a stone in the soil in the water	2	each correct answer = 1 mark Accept other suitable habitats shown in the picture

Question	Answer	Marks	Further Information
10(a)	magnesium + oxygen $\rightarrow$ magnesium oxide	1	Accept = instead of $\rightarrow$ either order for the reactants
10(b)	any two from fizzes / bubbles gets smaller / (eventually) disappears / forms a colourless solution yellow flame floats / moves on the surface solution left is alkaline	2	each correct answer = 1 mark Accept produces a gas / produces hydrogen Accept dissolves

Question	Answer	Marks	Further Information
11(a)	yes (no mark) and as the number of turns increases, the number of paper clips picked up increases	1	Ignore makes the electromagnet stronger
11(b)	repeat the experiment (for each number of turns) as this will improve the reliability of the experiment / so results can be checked / so averages can be taken or do the experiment with different number of turns, e.g. 1, 3, 5, 7 and/or 9 turns makes it easier to see any patterns / as this will provide more data or use smaller paper clips (idea that) more paper clips are picked up / improves accuracy	2	correct improvement = 1 mark correct explanation = 1 mark Do not accept makes the investigation more accurate Accept other suitable ways of improving reliability, accuracy or precision Ignore reference to fair testing

Question	Answer	Marks	Further Information
12(a)	core magnet	1	both answers correct for the mark
12(b)	compass	1	
12(c)	All arrows must be in the same direction 	1	Accept all arrows drawn in opposite direction if the learner assumes the Earth's magnetic north pole is the same as the geographical north pole
Question	Answer	Marks	Further Information
13(a)	three points correctly plotted within $\pm$ half a square = 1 mark curve of best fit through the points as judged by eye = 1 mark	2	Accept ecf from incorrectly plotted points Do not accept dot to dot / curve that goes through the origin
13(b)	increases / rises / is greater	1	

Question	Answer	Marks	Further Information
14(a)	A	1	more than one letter circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
14(b)	D	1	more than one letter circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
14(c)	B	1	more than one letter circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
Question	Answer	Marks	Further Information
15	force x distance is a moment balance when moments are equal / balance when clockwise moment equals anticlockwise moment but for 2 marks $F_1 \times d_1 = F_2 \times d_2$	2	Accept $F_1 \times d_1$ or $F_2 \times d_2$ answer must refer to moments rather than force or pressure

BLANK PAGE